

Lightweight Authentication and Communication Protocol for IoT Devices

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Abstract—At present, smart home based Internet of Things (IoT) technology has been developed and applied widely. Applications include automatic or remote control on-off electronics and electrical appliances in which users can send commands to a device gateway or an IoT device via smart devices. During the past few years, many research papers have investigated security and authentication protocols for IoT devices. However, they are still lack of consideration for the power constraints of IoT devices. This work presents a security and authentication protocol which reduces the use of hash functions and encryption method. Therefore, it consumes less power during the communication operation and, hence, complies with the power limitation of IoT and mobile devices.

Keywords—smart home, IoT, lightweight protocol, authentication protocol, security protocol

I. INTRODUCTION

Smart home is an application of IoT technology to control the electronic or electrical devices through the Internet. With the fast pace and leaps in technology development, the cost of these devices is getting cheaper and thus making these devices easier to acquire. They are widely used to monitor, control, manage and operate electrical and electronics home appliances [10].

Nowadays, most jobs rely on the Internet. Therefore, internet security must be taken into account to be able to prevent threats from outsiders or attackers who may obstruct or harm the smart home or even being harnessed to be used in attacks [11]. Studies have shown that smart homes still have many limitations, e.g. vulnerable to attack, low processing power and low power consumption. This limitation requires a need to present lightweight algorithms such as authentication methods in IoT to stop many kinds of attacks [12, 13, 14].

At present, the Internet of Things (IoT) technology has developed rapidly, resulting in the evolution of communication between users and home appliances. This will greatly accommodate user's daily life, such as real-time devices status checking or intuitive control of electrical appliances through smartphones or mobile devices. However, connecting to a wireless home network such as ZigBee, Zwave [1] and Wi-Fi may expose to a risk of confidentiality and security. It is possible that the message might be intercepted or spoofs from the smart home devices. Many smart home security researchers [2, 3, 4, 5] have presented a method for securing Mutual

Authentication, Confidentiality, and maintaining the integrity of the message. However, most works still lack consideration of the power consumption limitations of IoT devices [6,7,8, 9]. In the security aspect, they also lack the ability to prevent attacks by Eavesdropping. Therefore, we present the lightweight security and authentication of the smart home devices. Moreover, the proposed protocol consumes less power consumption compared to existing one. Our method uses only symmetric encryption and the reduced hash function while retaining the security features and maintains necessary security such as authentication, confidentiality and integrity. We also employ key distribution techniques to prevent duplicate key usage and duplicate messaging attacks.

II. RELATED WORK

Alshahrani and Traore [2] propose a secure lightweight mutual authentication and key exchange protocol for IoT smart home environment based on temporary identity and cumulative Keyed-hash chain. Lyu et al. [4] propose an efficient and anonymous authentication scheme for smart home environment using Elliptic Curve Cryptography (ECC) to avoid keeping the verification table for authentication purposes. Wazid et al. [5] propose a new secure remote user authentication scheme for a smart home environment. The scheme is efficient for resource-constrained smart devices with limited resources as it uses only one-way hash functions, bitwise XOR operations and symmetric encryptions/decryptions. Shuai et al. [6] propose an efficient and anonymous authentication scheme for smart home environment using Elliptic Curve Cryptography to avoid keeping the verification table for authentication purposes. Naoui et al. [7] propose a new secure and lightweight remote user authentication scheme for a smart home environment using Elliptic Curve Cryptography. However, these schemes, may be lightweight, but still contains a number of computation operations such as hash functions and encryptions and therefore consume more powers. We propose the protocol that will demonstrate these operations with less computational operations and hence low power consumption.

III. MATERIALS AND METHODS

We presents protocols to address existing research problems and limitations in IoT network security such as authentication, message integrity and message confidentiality and resistance to attacks such as intercepting messages or Eavesdropping Attack, Impersonation Attack, Man-in-the-

Middle Attack, Disguise or Masquerade Attack, Message Forgery Attack, Replay Attack, and power limitation.

The symbols used in our protocol are shown in Table I.

TABLE I. SYMBOLS USED IN THIS ARTICLE

Symbol	Meaning
RA	Registration Authorization
SP	Smart Phone
SHD,SD	Smart Home Device, Smart Device
ID_{SP}	Identity of SP
ID_{SHD}	Identity of SHD
SN_{SP}	Serial number of SP
SN_{SHD}	Serial number of SHD
MK	Master Key
SK	Session Key
$H(\bullet)$	Hash function
T	Time stamp
N	Nonce random number
$E\{\bullet\}$	Symmetric Encryption
ΔT	Maximum delay time for message flow
$A, A^*, B, B^*, D, D^*, E, E^*$	Hash values
C, F, M	Encrypt messages

The protocol that includes the Registration, Authentication and Session Key Distribution is demonstrates along with the connection and dataflow between SP, RA and SHD is shown Fig. 1.

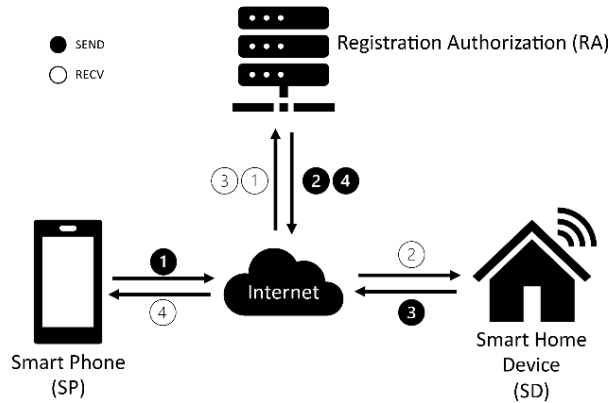


Fig. 1. Smart home connection system.

A. Register Phase

The purpose of registration is to share master keys between RA and SP or MK_{RA-SP} and between RA and SHD or MK_{RA-SHD} . Both keys sharing are performed over a secure channel such as

Secure Socket Layer (SSL) or Wireless Transport Layer Security (WTLS).

B. Authentication and Key Agreement Phase

The following steps describe the process of authentication and agreement.

- Step 1: $SP \rightarrow RA: \{ID_{SP}, ID_{SHD}, SN_{SP}, T1, N1, A\}$
Step 2: $RA \rightarrow SHD: \{ID_{SP}, ID_{SHD}, SN_{SP}, T2, N2, C\}$
Step 3: $SHD \rightarrow RA: \{SN_{SHD}, T3, N3, D\}$
Step 4: $RA \rightarrow SP: \{SN_{SHD}, T4, N4, F\}$
Step 5: $SP \rightarrow SHD: \{SK_{SP-SHD} \{Message_i, T_i\}\}$

Each step is elaborated in details as the following:

Step 1: $SP \rightarrow RA: \{ID_{SP}, ID_{SHD}, SN_{SP}, T1, N1, A\}$

- SP adds timestamp T1
- SP randoms nonce number N1
- SP computes hash value
 $A = H(ID_{SP}, ID_{SHD}, SN_{SP}, T1, N1, MK_{RA-SP})$
- SP sends message
 $\{ID_{SP}, ID_{SHD}, SN_{SP}, T1, N1, A\}$ to RA
- SP saves T1 and N1 in memory

This step is shown by the message flow from ① to ① in Fig. 1.

Step 2: $RA \rightarrow SHD: \{ID_{SP}, ID_{SHD}, SN_{SP}, T2, N2, C\}$

- RA adds current time stamp T2 and check delay time if $T2 - T1 < \Delta T$; RA will continue processing the message, otherwise, RA ends the communication with SP
- RA computes hash value A^* using $ID_{SP}, ID_{SHD}, SN_{SP}, T1$ and N1 from SP and use the share key $SHD(MK_{RA-SHD})$ from the registration phase, where
 $A^* = H(ID_{SP}, ID_{SHD}, SN_{SP}, T1, N1, MK_{RA-SP})$ and check the hash value. If $A^* = A$; RA will continue processing the message, otherwise, RA ends the communication with SP
- RA random nonce number N2
- RA assigns session key to communicate with SP and SHD (SK_{SP-SHD})
- RA computes hash value
 $B = H(ID_{SP}, ID_{SHD}, SN_{SP}, T2, N2, SK_{SP-SHD})$
- RA encrypts B and SK_{SP-SHD} with MK_{RA-SHD} to generate $C = EMK_{RA-SHD} \{SK_{SP-SHD}, B\}$
- RA sends message $\{ID_{SP}, ID_{SHD}, SN_{SP}, T2, N2, C\}$ to SHD
- RA saves $ID_{SP}, ID_{SHD}, SN_{SP}, T1, N1, SK_{SP-SHD}$ and C in memory

This step is shown by the message flow from ② to ② in Fig. 1.

Step 3: $SHD \rightarrow RA: \{SN_{SHD}, T3, N3, D\}$

- SHD receives message from RA and adds current time stamp T3 and check delay time if $T3 - T2 < \Delta T$; SHD will continue processing the

- message, otherwise, ends the communication with RA
- SHD decrypts message C with MK_{RA-SHD} to reveal SK_{SP-SHD} and B
 - SHD computes hash value $B^* = H(ID_{SP}, ID_{SHD}, SN_{SP}, T2, N2, SK_{SP-SHD})$ and compare if $B^* = B$; SHD will continue processing the message, otherwise, ends the communication with RA
 - SHD random nonce number N3
 - SHD computes hash value $D = H(SN_{SHD}, T3, N3, C, MK_{RA-SHD})$
 - SHD sends message $\{SN_{SHD}, T3, N3, D\}$ to RA

This step is shown by the message flow from ③ to ③ in Fig. 1.

Step 4: RA $\rightarrow$ SP: $\{SN_{SHD}, T4, N4, F\}$

- RA adds current time stamp T4 and check delay time if $T4 - T3 < \Delta T$; RA will continue processing the message, otherwise, ends the communication
- RA computes hash value $D^* = H(SN_{SHD}, T3, N3, C, MK_{RA-SHD})$ and compare if $D^* = D$; RA will continue processing the message, otherwise, ends the communication with SHD
- RA random nonce number N4
- RA computes hash value $E = H(ID_{SP}, ID_{SHD}, SN_{SP}, T1, N1, SN_{SHD}, T4, N4, MK_{RA-SP})$
- RA encrypts E and SK_{SP-SHD} with MK_{RA-SP} to generate $F = EMK_{RA-SP} \{SK_{SP-SHD}, E\}$
- RA sends message $\{SN_{SHD}, T4, N4, F\}$ to SP

This step is shown by the message flow from ④ to ④ in Fig. 1.

Step 5: SP $\rightarrow$ SHD: $\{SK_{SP-SHD} \{Message_i, T_i\}\}$

- SP adds current time stamp T_i and check delay time if $T_i - T4 < \Delta T$; SP will continue processing the message, otherwise, ends the communication
- SP decrypts message F with MK_{RA-SP} to reveal SK_{SP-SHD} and E
- SP computes hash value $E^* = H(ID_{SP}, ID_{SHD}, SN_{SP}, T1, N1, SN_{SHD}, T4, N4, MK_{RA-SP})$ and compare if $E^* = E$; SP will continue processing the message, otherwise, ends the communication
- SP prepares Message_i to be sent to SHD
- SP encrypts Message_i and T_i with SK_{SP-SHD} to generate $M = ESK_{SP-SHD} \{Message_i, T_i\}$
- SP sends message M or $\{SK_{SP-SHD} \{Message_i, T_i\}\}$ to SHD

Both SP and SHD use the same session key SK_{SP-SHD} to communicate. Once SK_{SP-SHD} is expired, new session key will be generated.

IV. ANALYSIS OF THE PROTOCOL

A. Immune to Eavesdropping

If an attacker intercepts the messages, our protocol offers the ability to prevent eavesdropping of the messages. Because each message is encrypted in every step, therefore, an attacker cannot decipher and read the message. Encryption of the proposed protocol ensures that the messages sent to other party are secure.

B. Impersonation Attack

If an attacker want to spoof a device with legitimate communication in order to steal the confidential messages between communicating devices. The proposed protocol can prevent spoofing of devices. Attacker cannot impersonate a device with legitimate communication because it needs the encrypted and authenticated master key. If it is successfully disguise as a device on either side, a matching master key is still required to decrypt the communication messages.

C. Man-in-the-Middle Attack

If an attacker is in the middle of communication between devices, it can intercept and collect the communication data between devices. However, it is unable to decrypt messages and read them because each message sends between devices is always encrypted.

D. Masquerade Attack

If an attacker is disguised as a device with legitimate communications in order to send or receive the confidential messages. The proposed protocol can prevent this type of spoofing through the authentication from both sides.

E. Message-Forgery

If an attacker fakes messages that communicate between legitimate devices, the proposed protocol matches a master key on both sides with a nonce value. It also uses a hash function to verify the authenticity of a message and hence make this an unsuccessful attack.

F. Mutual Authentication

The proposed protocol uses a master key, session key and hash functions which enable mutual authentication. It can prove who sends or receives the message. Mutual authentication cannot be forged and modified.

G. Replay Attack

In each step of sending a message, the protocol will always keep a record of a time stamp and a unique nonce value. It is impossible to retransmit an intercepted message; otherwise it is caught by the protocol.

H. Message Confidentiality

The protocol encrypts a message with a master key and a session key. Therefore, the message has the property of confidentiality. It can not be decrypted on the other side without a matching master key.

I. Message Integrity

The protocol encrypts the message with a hash function. This makes it possible to verify that messages sent to each

other are still intact, correct and not edited or changed on the way by an attacker.

J. Low Power Consumption

The entire protocol reduces the use of hash functions, and performs less computations. Therefore, it consumes less power which is suitable for smart home or IoT devices.

V. PROTOCOL PERFORMANCE ANALYSIS

The proposed protocol is designed with energy efficiency in mind and provides the security features that are resistant to attacks. The performance can be measured by counting number of encryption, hash function and the messages sends back and forth. The result is given in Table II and Table III.

TABLE II. COMPARE NUMBER OF OPERATIONS USED IN THE ENCRYPTION PROTOCOL

Methods	Symmetric Encryption	Asymmetric Encryption	Hash Function	Message
[2]	-	4	17	2
[4]	4	-	12	4
[5]	4	-	22	4
[6]	4	-	16	4
[7]	8	-	26	4
Our Protocol	3	-	8	5

TABLE III. COMPARE ENERGY CONSUMPTION IN ENCRYPTION PROTOCOL

Methods	AES (1.21 μ J)	RSA (546.5 μ J)	SHA1 (0.76 μ J)	Total (μ J)
[2]	-	2,186	12.92	2,198
[4]	4.84	-	9.12	13.96
[5]	4.84	-	16.72	21.56
[6]	4.84	-	12.16	17
[7]	-	-	19.76	19.76
Our Protocol	3.63	-	6.08	9.71

Table II and Table III show the number of encryption algorithm used and energy consumption in the encryption protocol comparing with existing research. It has been shown that our protocol takes less computational operations and consumes low energy than all other protocols in [2, 4, 5, 6, 7]. Therefore, it is an efficient protocol with low power consumption according to power calculation in cryptography [15, 16].

VI. PROTOCOL VALIDATION WITH THE SCYTHYR TOOL

The proposed protocol uses the Scyther Tool [17] to verify the validity and security of the protocol. The validation code is shown in Fig. 2 and the output from executing the code is shown in Fig. 3. The OK status of all claims is Alive, Weakagree, Niagree, and Nisynch, which indicate that the proposed protocol is valid and resistant to attacks.

```

usertype String;
usertype TimeStamp;
usertype SessionKey;
hashfunction H;
const IDSP,IDRA,IDSHD,SNSP,SNSHD: String;
const MKRA-SP,MKRA-SHD,SKSP-SHD: SessionKey;
macro M1 =IDSP,IDSHD,SNSP,T1,N1,H(IDSP,IDSHD,SNSP,T1,N1,MKRA-SP);
macro M2 =IDSP,IDSHD,SNSP,T2,N,{SKSPSHD,

```

```

H(IDSP,IDSHD,SNSP,T2,N2,SKSP-SHD)}MKRA-SHD;
macro M3 = SNSHD,T3,N3,H(SNSHD,T3,N3),{SKSPSHD,
H(IDSP,IDSHD,SNSP,T2,N2,SKSP-SHD)}MKRA-SHD;
macro M4 = SNSHD,T4,N4,H(SNSHD,T4,N4),{SKSPSHD,
H(IDSP,IDSHD,SNSP,T1,N1,SNSHD,T4,N4,MKRA-SP)}MKRA-SP;
protocol AUKA(SP,RA,SHD) {
  role SP {
    fresh T1,T4: TimeStamp;
    fresh N1,N4: Nonce;
    send_1(SP,RA,M1);
    recv_4(RA,SP,M4);
    claim_SP1(SP,Alive);
    claim_SP2(SP,Weakagree);
    claim_SP3(SP,Niagree);
    claim_SP4(SP,Nisynch)
  }
  role RA {
    fresh T1,T2,T3,T4: TimeStamp;
    fresh N1,N2,N3,N4: Nonce;
    recv_1(SP,RA,M1);
    send_2(RA,SHD,M2);
    recv_3(SHD,RA,M3);
    send_4(RA,SP,M4);
    claim_RA1(RA,Alive);
    claim_RA2(RA,Weakagree);
    claim_RA3(RA,Niagree);
    claim_RA4(RA,Nisynch);
  }
  role SHD {
    fresh T2,T3: TimeStamp;
    fresh N2,N3: Nonce;
    recv_2(RA,SHD,M2);
    send_3(SHD,RA,M3);
    claim_SHD1(SHD,Alive);
    claim_SHD2(SHD,Weakagree);
    claim_SHD3(SHD,Niagree);
    claim_SHD4(SHD,Nisynch);
  }
}

```

Fig. 2. Validation Codes in Scyther Tool

Scyther results : verify

Claim		Status	Commer			
AUKA	SP	AUKA,SP1	Alive	Ok	Verified	No attacks.
		AUKA,SP2	Weakagree	Ok	Verified	No attacks.
		AUKA,SP3	Niagree	Ok	Verified	No attacks.
		AUKA,SP4	Nisynch	Ok	Verified	No attacks.
RA		AUKA,RA1	Alive	Ok	Verified	No attacks.
		AUKA,RA2	Weakagree	Ok	Verified	No attacks.
		AUKA,RA3	Niagree	Ok	Verified	No attacks.
		AUKA,RA4	Nisynch	Ok	Verified	No attacks.
SD		AUKA,SD1	Alive	Ok	Verified	No attacks.
		AUKA,SD2	Weakagree	Ok	Verified	No attacks.
		AUKA,SD3	Niagree	Ok	Verified	No attacks.
		AUKA,SD4	Nisynch	Ok	Verified	No attacks.

Done.

Fig. 3. Test and Validation result.

VII. CONCLUSION

This research presents an authentication protocol. It adopts session key propagation and hash function to create a lightweight protocol with less computation that make it suitable for smart home and wireless devices. It also contains the message authentication, confidentiality and integrity. In addition, the proposed protocol consumes less power compared to existing protocols. Finally, the validity of the protocol is verified with the Scyther Tool to ensure that the protocol was secure and can be employed for future research and application.

REFERENCES

- [1] XX.C. Gomez, and J. Paradells, "Wireless home automation networks: A survey of architectures and technologies," *IEEE Communications Magazine*, vol. 48, no. 6, pp. 92–101, 2010.
- [2] M. Alshahrani, and I. Traore, "Secure mutual authentication and automated access control for IoT smart home using cumulative Keyed-hash chain," *Journal of Information Security and Applications*, vol. 45, pp. 156–175, 2019.
- [3] P. Kumar, A. Braeken, A. Gurtov, J. Iinatti, and P.H. Ha, "Anonymous Secure Framework in Connected Smart Home Environments," in *IEEE Transactions on Information Forensics and Security*, vol. 12, no. 4, pp. 968–979, 2017.
- [4] Q. Lyu, N. Zheng, H. Liu, C. Gao, S. Chen, and J. Liu, "Remotely Access My Smart Home in Private: An Anti- Tracking Authentication and Key Agreement Scheme," in *IEEE Access*, vol. 7, pp. 41835–41851, 2019.
- [5] M. Wazid, A.K. Das, V. Odelu, N. Kumar, and W. Susilo, "Secure Remote User Authenticated Key Establishment Protocol for Smart Home Environment," in *IEEE Transactions on Dependable and Secure Computing*, vol. 17, no. 2, pp. 391–406, 2020.
- [6] M. Shuai, N. Yu, H. Wang, and L. Xiong, "Anonymous authentication scheme for smart home environment with provable security," *Computers & Security*, vol. 86, pp. 132–46, 2019.
- [7] S. Naoui, M. E. Elhdhili, and L.A. Saidane, "Lightweight and Secure Password Based Smart Home Authentication Protocol: LSP-SHAP," *Journal of Network and Systems Management*, 2019.
- [8] S. Bhardwaj, T. Ozcelebi, J. Lukkien, and C. Uysal, "Resource and service management architecture of a low capacity network for smart spaces," *Consumer Electronics, IEEE Transactions on*, vol. 58, no. 2, pp. 389–396, 2012.
- [9] P. Kumar, A. Gurtov, J. Iinatti, M. Ylianttila, and M. Sain, "Lightweight and Secure Session-Key Establishment Scheme in Smart Home Environments," in *IEEE Sensors Journal*, vol. 16, no. 1, pp. 254–264, 2016.
- [10] S.S.I. Samuel, "A review of connectivity challenges in IoT-smart home," 2016 3rd MEC International Conference on Big Data and Smart City (ICBDSC), Muscat, Oman, pp. 1–4, 2016.
- [11] C. Kolias, G. Kambourakis, A. Stavrou, and J. Voas, "DDoS in the IoT: Mirai and Other Botnets," in *Computer*, vol. 50, no. 7, pp. 80–84, 2017.
- [12] M. Yu, J. Zhuge, M. Cao, Z. Shi, and L. Jiang, "A Survey of Security Vulnerability Analysis, Discovery, JIST Journal of Information Science and Technology Volume 12, NO 2 | JUL – DEC 2022 | 21-30 30 Detection, and Mitigation on IoT Devices." *Future Internet*, 12, 27, 2020.
- [13] A. Hameed and A. Alomary, "Security Issues in IoT: A Survey," 2019 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT), Sakhier, Bahrain, 2019, pp. 1–5.
- [14] F.A. Alaba, M. Othman, I.A.T. Hashem, and F. Alotaibi, "Internet of Things security: A survey," *Journal of Network and Computer Applications*, vol. 88, pp. 10–28, 2017.
- [15] N.R. Potlapally, S. Ravi, A. Raghunathan, and N. K. Jha, "A study of the energy consumption characteristics of cryptographic algorithms and security protocols," in *IEEE Trans. Mobile computing*, vol. 5, no. 2, pp. 128–143, 2006.
- [16] A. Castiglione, F. Palmieri, U. Fiore, A. Castiglione, and A. De Santis, "Modeling energy-efficient secure communications in multi-mode wireless mobile devices," *Journal of Computer and System Sciences*, vol. 81, no. 8, pp. 1464–1478, 2015.
- [17] C.J.F. Cremers, "The scyther tool: verification, falsification, and analysis of security protocols," In: *International Conference on Computer Aided Verification*, Springer, pp. 414–418, 2008.